Automotive Technical Service Diagnostic Bulletin

Document ID: DIAG-C-CLASS-P0300
Classification: OEM Level 3 Diagnostic & Repair Procedure
Target Vehicle: Mercedes-Benz C-Class (W206 / C 300)
Powertrain: 2.0L Turbocharged Inline-4 (M254) with 2nd Gen ISG (Beltless 48V Mild-Hybrid)
Engine Control System: Bosch MD1CS006 (Engine Control Module N3/10) - High-Speed CAN-E2 Architecture
Electrical Architecture: Integrated 12V / 48V Architecture with 48V DC/DC Converter (N83/1)
Fuel Injection Architecture: Single Cam-Driven High Pressure Pump (HPP) with Solenoid Direct Injectors (200-250 bar)
Firing Order: 1-3-4-2

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1. DIAGNOSTIC TROUBLE CODE (DTC) SPECIFICATION

- **DTC Code:** `P0300`
- **Standard Fault Description:** Random / Multiple Cylinder Misfire Detected
- **System Definition:** Engine control module detects crankshaft acceleration irregularities across multiple random cylinders exceeding the emissions/catalyst damage threshold.
- **MIL (Malfunction Indicator Lamp) Status:** Illuminated (Continuous or Flashing depending on catalyst severity)
- **Monitoring Condition:** Continuous monitoring during engine running cycle with coolant temp > 60°C and vehicle speed > 0 km/h.

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2. PLATFORM-SPECIFIC ROOT CAUSE ANALYSIS (RCA)

Root cause analysis customized specifically for the **Mercedes-Benz C-Class (W206 / C 300)**:

1. **48V Integrated Starter Generator (ISG) Beltless Damper Torsional Resonance:** Micro-vibrations from 48V motor generator (M70) dampening algorithm causing phantom CKP sensor tooth timing deceleration.
2. **Low-Pressure Fuel Supply Cavitation:** Electric in-tank fuel delivery module (M3) fuel filter strainer partially clogged with tank debris, starving high-pressure pump (HPP) under heavy throttle.
3. **Direct-Injection Solenoid Injector High-Pressure Rail Resonator Leak:** Pressure drop across the compact 4-cylinder single fuel rail assembly.

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3. DEDICATED WIRING DIAGRAM & CIRCUIT PINOUT

Schematic configuration and electrical pin connections specifically for the C-Class control harness:

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[N3/10 ECM (Bosch MD1CS)]              [Direct Ignition Coils T1/1 - T1/4]
Pin B15 (Cyl 1 Spark Trigger) --------> Pin 3 (Signal 5V) [T1/1]
Pin B16 (Cyl 2 Spark Trigger) --------> Pin 3 (Signal 5V) [T1/2]
Pin B17 (Cyl 3 Spark Trigger) --------> Pin 3 (Signal 5V) [T1/3]
Pin B18 (Cyl 4 Spark Trigger) --------> Pin 3 (Signal 5V) [T1/4]
Circuit 15 Protected (Fuse F32/1-F12) -> Pin 1 (+12V Switched Power)
Engine Block Ground W10 --------------> Pin 2 (Coil Ground)
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- Pinout Voltage Test: Pin 1 = 12.2V - 14.5V; Pin 2 to Block < 0.1Ω; Pin 3 = 5V logic pulses during engine cranking.

Electrical Reference Points

- **Primary Ground Reference:** `W10 (Engine Block Ground Point, Right) & W9 (Front Wheel Arch Ground)`
- **Power Distribution Node:** `Engine Bay Prefuse Box F32/1 & Front SAM Module (N10/1)`
- **Signal Logic Voltage Range:** 0.0V to 5.0V regulated DC (unless high-voltage direct injection rail)

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4. STEP-BY-STEP DIAGNOSTIC & VERIFICATION PROCEDURE

Follow this OEM workshop sequence to isolate the failure on this platform:

1. Connect XENTRY Diagnostic System; verify if DTC P0300 is accompanied by U0111 (Lost Comm with ISG N129).
2. Check low pressure fuel system pressure via B4/1 sensor (Spec: 5.5 bar to 6.5 bar constant).
3. Perform high-pressure leak-down test: turn engine off, watch rail pressure; must not drop more than 5 bar in 10 minutes.
4. Check ground point W10 on right side engine block for correct torque (10 Nm).

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5. PRECISE RESOLUTION & REPAIR PROCEDURE

- Update software calibration on N129 (Starter-Generator Control Unit) and N3/10 (Engine Control Unit) to resolve crankshaft torsional harmonic misdetection.
- Clean ground stud W10 and coat with dielectric protective grease.
- Flush fuel tank and replace in-tank low-pressure delivery pump module M3 if supply pressure drops under 4.8 bar.

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6. POST-REPAIR QUALITY VALIDATION & RE-TESTING

1. Clear all DTCs from Engine Control Module (`Bosch`) and secondary gateway modules.
2. Perform comprehensive drive cycle:
3. Cold start idle for 3 minutes (monitors secondary air / catalyst preheat and cold misfire).
4. Accelerate smoothly to 50 km/h and hold for 2 minutes.
5. Cruise at 80–100 km/h on highway for 10 minutes (validates downstream O2 sensor and fuel trim closed-loop balance).
6. Coasting deceleration (fuel cut-off check) for 5 seconds without braking.
7. Re-scan control units with diagnostic tool; confirm all OBD-II readiness monitors show **READY** / **COMPLETE** and no pending DTCs are stored.